Frequency Decoupling Fusion for Image Super-Resolution of Remote Sensing

Kanghui Zhao, Tao Lu, Jiaming Wang, Yu Wang, Yuanzhi Wang, and Yanduo Zhang

***Abstract*—Benefiting from the excellent global expression ability, transformer-based image super-resolution (SR) has made significant progress. However, the existing transformer-based SR methods still have the problem of high-frequency information reconstruction loss when processing remote sensing images due to their wide imaging range, rich high-frequency information, and large differences, which affects the characterization ability of the transformer. In addition, the high computational overhead is unacceptable. To alleviate the above problems, we consider the remote sensing image SR from the perspective of the frequency domain. Specifically, we propose an efficient frequency decoupling-fusion framework (FDFNet), which is called FDFNet. In particular, we consider that when a large amount of previous work was carried out to extract features in the spatial domain, it was very easy to lose the high-frequency information in the original image. Therefore, we first introduce a frequency decoupling block (FDB), which decouples the image into low- and high-frequency components, processes high- and low-frequency information, respectively, in a divide-and-conquer manner, and restores high-frequency details before delving deeper. Furthermore, we notice that spatial self-attention is a low-pass filter that tends to have global perception and to demonstrate limitations in reconstructing high-frequency details. Therefore, we meticulously designed a parallel frequency-aware transformer module (PFTM) to extract spatial frequency attention and channel transposition attention, which enables our model to focus more on local texture details to restore high-frequency details. A large number of experimental results on multiple public datasets show that our FDFNet outperforms the state-of-the-art (SOTA) SR methods in quantitative metrics and visual quality, and achieves a balance between performance and efficiency within a limited computing budget.**

***Index Terms*—Dual-frequency aggregation, frequency decoupling fusion, remote sensing image, super-resolution (SR).**

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Kanghui Zhao, Tao Lu, Jiaming Wang, and Yanduo Zhang are with Hubei Key Laboratory of Intelligent Robot, Wuhan Institute of Technology, Wuhan 430205, China (e-mail: zhaokanghui1024@gmail.com; lutxyl@gmail.com; wjmecho@wit.edu.cn; zhangyanduo@hotmail.com).

Yu Wang is with the State Key Laboratory for Information Engineering in Surveying, Mapping and Remote Sensing, Wuhan University, Wuhan 430079, China (e-mail: wy2022@whu.edu.cn).

Yuanzhi Wang is with the School of Computer Science and Engineering, Nanjing University of Science and Technology, Nanjing 210094, China (e-mail: yuanzhiwang@njust.edu.cn).

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I. Introduction

HIGH-QUALITY remote sensing images have been widely used in many large-scale fine-grained fields such as meteorological observation, resource mapping, disaster monitoring, military target detection, and scene classification [1], [2], [3], [4], [5]. However, due to hardware and environmental factors, the resolution of the acquired remote sensing images is usually low, and there is a problem of detail loss. In view of the high cost of improving remote sensing image quality from the aspect of hardware, image super-resolution (SR) reconstruction technology has become an effective solution to improve the resolution and quality of remote sensing images.

Image SR that aims to learn prior knowledge from a low-resolution (LR) image and reconstruct a potentially high-resolution (HR) image has been widely used in remote sensing. Recently, various remote sensing image SR methods based on convolutional neural networks (CNNs) [6], [7], [8], [9], [10] have been developed and achieved excellent performance. However, due to well-known reasons, the local perceptual characteristics and limited receptive field of CNNs limit their global representation ability, which leads to suboptimal reconstruction results. This requirement is essential for SR tasks, especially in remote sensing images with a wider imaging range, as the predicted pixels require SR from prior knowledge of their surrounding region.

In recent years, transformer has outperformed CNN in terms of image SR because it better models image structure globally by using a self-attention mechanism, such as HAT [11], HAUNet [12], and TTST [13]. HAUNet [12] proposed a hybrid attention mechanism for remote sensing image SR to efficiently explore multiscale features and enhance global feature representation through hybrid convolution-based attention. HAT [11] combines channel attention and window-based self-attention, while taking advantage of the complementary advantages of global statistics and strong local fitting capabilities, and introduces overlapping cross-attention modules to enhance the interaction between adjacent window features. TTST [13] designed a top-k token-selective transformer to reconstruct HR images and developed a residual token selection group to select the most important features. However, these methods are forced to limit the attention calculation to the local window, thus restricting the receptive field and affecting the performance. Despite the attractive performance of these models, the following challenges remain in their application in remote sensing SR.

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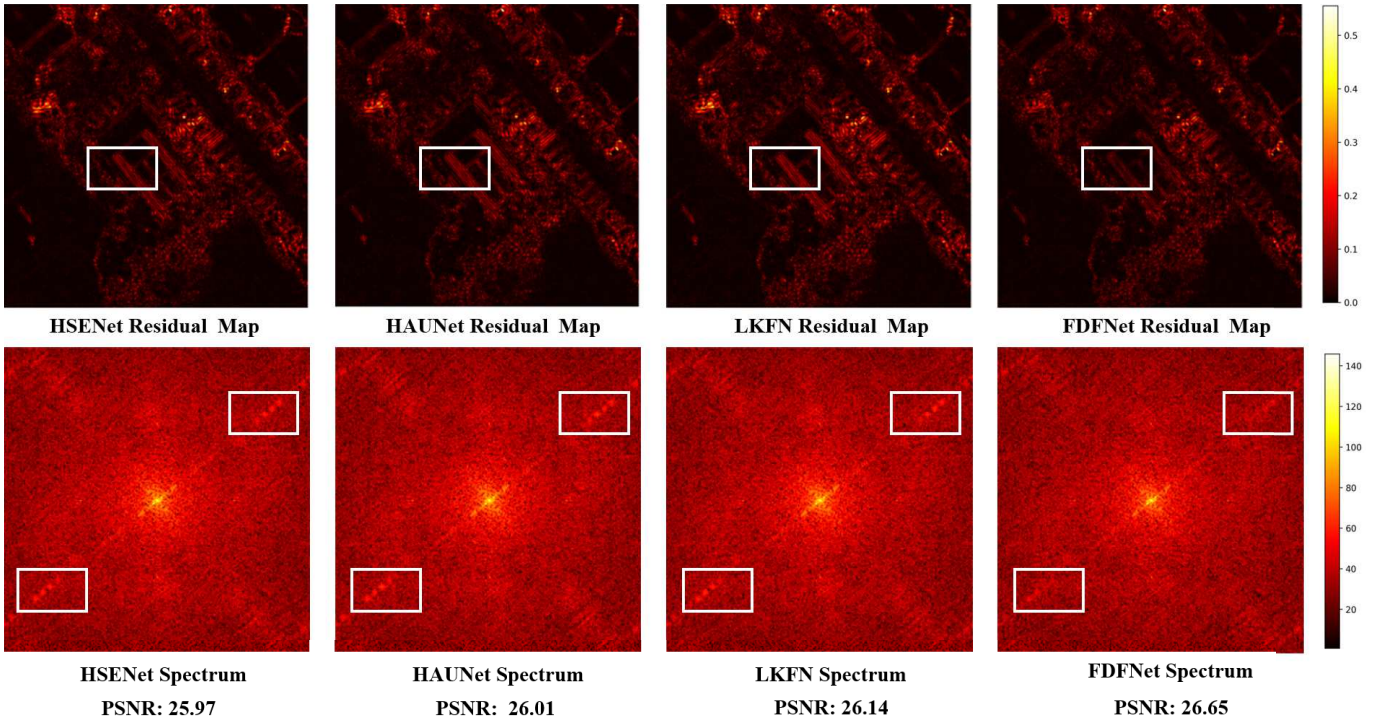


Fig. 1. Comparative analysis of SR results. The CNN- and transformer-based method (HSENet, LKFN, and HAUNet) produces visually smoother outputs, and its error residual exhibits stronger high-frequency energy (brighter periphery in the spectrum). This indicates a tendency for transformer architectures to lose high-frequency details during reconstruction. On the contrary, the error residuals of our method exhibit weaker high-frequency energy (the darker periphery in the spectrum, as indicated by the white box in the figure).

1) *Computational Complexity:* At present, most remote sensing SR methods based on transformer perform self-attention calculation on pixels, resulting in increased model calculation amount. Other methods limit attention calculation to local windows, constraining the sensitivity field and affecting reconstruction performance.
2) *High-Frequency Information Loss:* Remote sensing images have the characteristics of wide imaging, rich texture frequency information, and a large difference in object scale. Most transformer-based SR methods use a small window size for self-attention computation, which makes the model more inclined to retain low-frequency information, while ignoring the reconstruction of rich high-frequency information, thus hindering the transformer's presentation capability. As Park and Kim [14] described, the existing vision transformer is a low-pass filter.

To empirically validate this limitation, we conducted a preliminary experiment (as shown in Fig. 1). We compared the frequency-domain error spectra of a leading transformer method (HAUNet [12] and LKFN [15]) and a CNN-based method (HSENet [6]). Qualitatively speaking, although the three models may perform similarly in overall metrics such as peak signal-to-noise ratio (PSNR), the reconstruction error of the transformer is significantly concentrated in the high-frequency band, which confirms its inherent low-frequency reconstruction bias and weaknesses in restoring fine textures and edges.

Based on the above considerations, a natural question arises: How to reduce the model size while recovering the high-frequency image detail of the transformer-based remote sensing SR approach?

Patro and Agneeswaran [16] alleviated the problem of high-frequency information loss associated with downsampling operations by separating the low- and high-frequency components. Meanwhile, this method uses Einstein's multiplication to mix tokens and channels, effectively reducing complexity through the spectrum-gated network. This inspires us to shift from the existing spatial self-attention perspective to the frequency perspective. Inspired by the above observations, we propose a frequency decoupling-fusion framework (FDFNet), a frequency-aware transformer for lightweight remote sensing image SR. Our FDFNet mainly consists of a frequency decoupling block (FDB) and a parallel frequency-aware transformer module (PFTM), which reconstructs high frequencies through the interaction of spatial, channel, and frequency information. Specifically, FDB first decomposes the input image into high- and low-frequency components for processing and, finally, aggregates and restores high-frequency details, which are simultaneously used as the input for the next stage of the parallel frequency-aware attention module (PFTM). Subsequently, PFTM fuses the high-frequency information extracted from the shallow features with the channel information and further enhances the high-frequency components in the shallow features. To further enhance the feature representation capability, PFTM also integrates spatial frequency attention and channel transposition attention mechanisms. Through this

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design, FDFNet can effectively capture spatial, frequency, and channel context information, promote cross-dimensional interblock feature aggregation, and make up for the deficiency of the attention mechanism in long-distance detail information capture. In addition, to further integrate the frequency-domain, spatial, and channel feature representations, we introduce a dual-frequency aggregation feedforward network (DAFN) and a frequency gating mechanism in the middle of the fully connected layer to aggregate frequency information. Overall, our FDFNet not only retains the global low-frequency information perception capability but also enhances the recovery ability of high-frequency details. This paper's primary contributions are articulated in the following.

1) We propose a novel paradigm for image frequency processing that decouples an image into low- and high-frequency components via a divide-and-conquer strategy. This approach facilitates independent enhancement and achieves high-frequency recovery along with multifeature aggregation through a PFTM of our design.
2) A computationally efficient and high-performance FDFNet is proposed. By effectively integrating spatial, channel, and frequency information, the framework reconstructs high-frequency representations within a transformer architecture, enabling high-quality reconstruction of LR remote sensing images. It not only preserves structural integrity and texture details effectively but also exhibits lower computational complexity and a more compact model size compared to state-of-the-art (SOTA) methods.
3) We performed SR reconstruction experiments on three public datasets and evaluated the results in conjunction with high-level vision tasks, such as semantic segmentation. The proposed FDFNet achieves SOTA performance among deep learning-based SR methods, outperforming existing approaches on both quantitative metrics and qualitative visual quality.

II. Related Work

A. Remote Sensing Image SR

Remote sensing image SR presents unique challenges compared to natural images, necessitating specialized network architectures to address their large spatial coverage. Some classic early CNN-based methods enhance local details to improve the performance of SR reconstruction through residual connections [17], dense connections [18], attention mechanisms [8], [19], and so on. However, this type of method is more inclined to infer the potential low-frequency information in LR images, which leads to the reconstruction results often being smoother and lacking texture details. To alleviate this issue, some studies have improved the restoration process of texture details by mining the display prior information of the image and integrating it into the reconstruction results. Zhao et al. [10], [20] extracted edge image from LR and constructed edge enhancement branches, fusing the SR reconstruction results with the edge reconstruction results to improve the structure and texture of the image. However, these methods obviously bring additional computational overhead. While significant progress has been made in CNN-based SR, these methods are nevertheless limited by their receptive field. This presents a particular challenge for remote sensing images, as their extensive spatial coverage, rich textual detail, and large-scale variations demand a stronger global perception capability than CNNs can typically provide.

The necessity to capture remote dependencies and augment the model's capacity for global image perception has driven increased research interest in transformer architectures, prompting the development of numerous related methods. Lei et al. [21] introduced a multistage enhanced transformer architecture that employs multiscale windows to balance global and local information, achieving effective fusion of multiscale high- and low-dimensional features. Chen et al. [11] developed a model that combines channel and window self-attention mechanisms to exploit their complementary strengths, augmented by an overlapping cross-attention module for richer interwindow interactions and the same-task pretraining strategy to maximize performance. To mitigate computational complexity, ESTNet [22] integrates channel and group attention mechanisms. It operates through a three-stage pipeline—shallow feature extraction, deep residual feature enhancement, and image reconstruction—to enhance reconstruction quality while maintaining a low computational cost. Meanwhile, TTST [13] employs a top-K token selection mechanism to dynamically prune redundant tokens. It further incorporates multiscale feedforward layers and global context attention, which significantly improves feature representation and reconstruction accuracy.

Compared to CNNs, transformer-based methods offer superior accuracy and enhanced perceptual capabilities, yet they continue to struggle with balancing efficiency and scalability. As a result, several lightweight model designs have emerged to achieve a better tradeoff between computational efficiency and performance.

B. Lightweight Architecture

According to [23], the development of lightweight SR methods has gone through three main stages: initially focusing on efficient upsampling techniques [24], then evolving into parameter-sharing strategies [25], and, finally, evolving into multistage distillation methods [26]. Subsequently, the attention mechanism was introduced to achieve a balance between local and global information [27]. A series of targeted innovations have also emerged in the field of remote sensing image SR, such as the channel segmentation attention mechanism proposed by Wang et al. [28] to enhance feature interaction and the context transformation network to reduce the number of parameters [9]. Representative works on hybrid architectures include the transformer-CNN fusion model proposed by Liang et al. [29], followed by improved mechanisms such as covariance-based attention [30] and permutation self-attention [31]. Binary neural networks (BNNs) have further significantly enhanced efficiency through adaptive binary convolution [32]. Recently, Zhi et al. [33] combined the state space model (Mamba) with the transformer and introduced auxiliary branches to better preserve detailed information. Overall, lightweight SR technology has evolved from the

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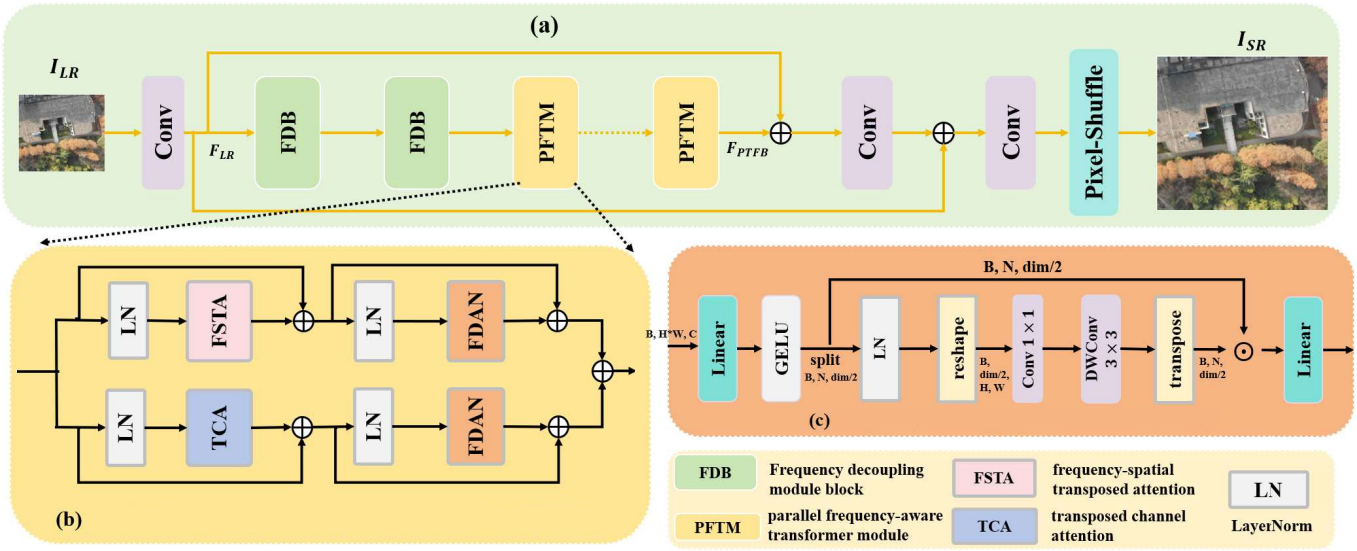


Fig. 2. Our FDFNet architecture primarily consists of two key components: the FDB and the PFTM. (a) Framework of FDFNet. (b) PFTM. (c) FDAN.

initial efficient CNN structure to hybrid architectures and emerging designs based on Mamba, continuously improving in terms of scalability and detail retention capabilities.

However, as previously discussed, remote sensing imagery exhibits significant variations in the scale of ground objects, encompassing everything from large infrastructures such as airports to much smaller elements such as individual vehicles. While most existing transformer-based models rely on spatial self-attention to capture feature dependencies—enabling them to integrate global contextual information—this approach also predisposes the model toward reconstructing low-frequency components, often at the expense of high-frequency details [14], [34].

C. Frequency-Guided Image Enhancement

Fast Fourier transform (FFT) can extract the global frequency-domain statistical features of images and, thus, is widely used in frequency-domain modeling in various visual tasks. In low-light visual tasks, FFT also demonstrates its significant role. Mao et al. [35] proposed a residual FFT module that can effectively capture local high-frequency details in images, thereby improving the enhancement effect of low-light images. Li et al. [36] embedded the Fourier transform into the deep network structure to suppress the common noise amplification problem during the luminance enhancement process. Guo et al. [37] further proposed a window-based frequency channel attention mechanism to enhance the model's sensitivity to frequency-domain features. In addition, Wang et al. [38] achieved complementary learning in face SR reconstruction by jointly utilizing frequency- and spatial-domain information. There are also some methods to introduce FFT into the loss function to enhance the recovery of high-frequency details in the reconstructed image [39]. Xiao et al. [40] adopted a lightweight dynamic filtering strategy, differentiating and weighting the frequency-domain features through the weights learned by the convolutional layer, and adaptively strengthening the information frequencies related to SR, thereby significantly enhancing the flexibility and accuracy of frequency selection and modulation.

However, existing methods often focus on designing complex network structures to mine frequency information but neglect the analysis of the contributions of different frequency components. This approach not only may amplify harmful frequency bands (such as high-frequency noise) but also brings additional computational burdens.

III. Our Method

The architecture of our proposed FDFNet is illustrated in Fig. 2 and consists of three primary components: a shallow feature extraction module, a deep feature extraction module, and an HR image reconstruction module. The process begins by extracting shallow features $F_{\mathrm{LR}} \in \mathbb{R}^{H\times W\times C}$ from the LR image I_{LR} input using a 3×3 convolutional layer. Subsequently, F_{LR} is sent to the proposed FDB, where high- and low-frequency information is processed, respectively, using a divide-and-conquer strategy. Next, multiple stacked PFTMs are used, combined with convolution operations and skip connections, to further refine the deep feature F_{PFTM}. In the final stage, the high-quality image I_{SR} is reconstructed by upsampling the feature F_{PFTM} using the pixel-shuffle operation.

A. Frequency Decoupling Block

As noted in prior work [14], feature extraction operating solely in the spatial-domain frequently fails to preserve high-frequency components of the original image. To mitigate this limitation, we introduce an FDB designed to process high- and low-frequency information within the shallow feature F_{LR} separately, thereby recovering otherwise lost high-frequency details before the subsequent deep features processing stage. Drawing inspiration from [16] and [34], we employ the dual-tree complex wavelet transform (DTCWT) to decompose FS in the frequency domain, enabling effective separation and independent handling of different frequency components.

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Mathematically, the formulation for the real component of the DTCWT output is

$$\mathbf{X}_F(u,v) = \mathbf{X}_\phi(u,v) + \mathbf{X}_\psi(u,v) \tag{1}$$

$$\mathbf{X}_\phi(u,v) = \sum_{h=0}^{H-1}\sum_{w=0}^{W-1} c_{M,h,w}\phi_{M,h,w} \tag{2}$$

$$\mathbf{X}_\psi(u,v) = \sum_{m=0}^{M-1}\sum_{h=0}^{H-1}\sum_{w=0}^{W-1}\sum_{k=0}^{6} d^k_{m,h,w}\psi^k_{m,h,w} \tag{3}$$

where $\mathbf{X}_\phi(u,v)$ and $\mathbf{X}_\psi(u,v)$ represent the low-frequency scaling and high-frequency wavelet components, respectively, while M and k denote the resolution and directional selectivity.

Correspondingly, the imaginary part of the DTCWT can be derived to obtain both $L_{\text{imaginary}}$ and $H_{\text{imaginary}}$. Here, $L_{\text{imaginary}} \in \mathbb{R}^{H\times W\times C}$ corresponds to the low-frequency component of the image. This component is subsequently processed using a combination of 1×1 and depthwise convolutions to integrate global low-frequency context with local detail, yielding the refined output $L_{\text{out}} \in \mathbb{R}^{H\times W\times C}$

However, extracting informative features from the high-frequency components $H_{\text{imaginary}} \in \mathbb{R}^{k\times H\times W\times C\times 2}$ presents a substantial challenge, owing to the increased dimensionality by a factor of k (from multiple directional angles) and the doubled parameter count resulting from the inclusion of both real and imaginary parts. To tackle this, we employ fully connected layers to integrate cross-channel and high-frequency information. For computational efficiency, the channels are partitioned into C_b blocks, each containing C_d channels, resulting in the reshaped representation $H_{\text{imaginary}} \in \mathbb{R}^{k\times H\times W\times C\times 2\times C_d\times C_b}$. Along the last dimension, two linear transformations are applied to the high-frequency features to capture fine-grained details such as edges, textures, and subtle patterns. Subsequently, an inverse transformation is applied to convert the spectrum features back to the physical domain, yielding the output $F_{\text{FDB}} \in \mathbb{R}^{H\times W\times C}$ Furthermore, residual connections are utilized to propagate the high-frequency information extracted by the F_{FDB} module via DTCWT across different depths of the network, enhancing feature reuse and integration.

In this study, we chose DTCWT based on its specific advantages in the SR task of remote sensing images. We chose DTCWT over the standard DWT mainly based on its two key advantages.

1) *Approximate Translation Invariance:* DTCWT is less sensitive to the translation of images, which helps improve the stability of SR reconstruction and reduce artifacts.
2) *Higher Directional Selectivity:* DTCWT captures details in six directions (while DWT only has three), which can more effectively represent the complex edge and texture patterns commonly found in remote sensing images, thereby more accurately separating high-frequency information.

We visualize the results of FDB, as shown in Fig. 3. It should be noted first that, to avoid interference from training parameters, the pretrained model was not loaded in the output results. From left to bottom in this figure, the basic frequency splitting process of FDB for LR input images is fully displayed. The physical meanings and functional verifications of each part (combined with the characteristics of the untrained state) are given as follows.

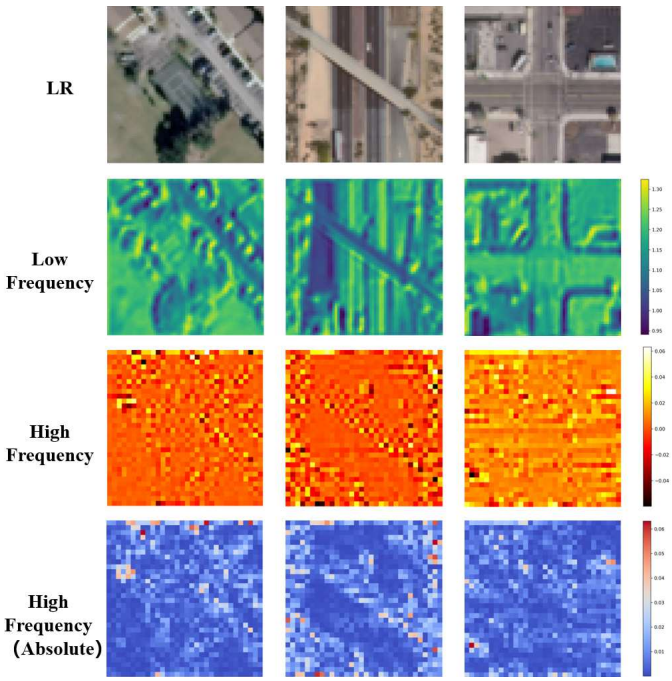


Fig. 3. Visualization of intermediate outputs from the FDB without pretrained models. (From top to bottom) LR remote sensing images, decomposed low-frequency components (capturing global scene structures), high-frequency components (retaining phase information of fine textures and edges), and absolute values of high-frequency components (highlighting the magnitude distribution of detailed features).

The low-frequency component is obtained by the decomposition of FDB through DTCWT, mainly carrying the global smooth structure of the image and the information of large-area uniform regions (such as the large-scale terrain base in the image, the main contour of large buildings, and continuous site boundaries). It can be seen from this that even without a pretrained model, this component can still effectively strip away the interference of high-frequency details in the input and independently learn the low-frequency core of the overall image shape.

The high-frequency component is obtained through the multidirectional wavelet decomposition of DTCWT. The core contains the detailed texture, edge features, and local gray-level mutation signals of the image and strictly retains the phase polarity of the high-frequency signal (manifested as the distribution of positive and negative gray-level values in the image), for example, the gradient transition direction from the bright area to the dark area, and the alternating light and dark rules of the texture. Since the pretrained model has not been loaded, this component may contain a small amount of residual low-frequency structure and weak noise. However, the key high-frequency information, such as edges and textures that were originally superimposed with the low-frequency structure in the LR input, is still separated to avoid the problem of high-frequency information being submerged in traditional spatial-domain feature extraction.

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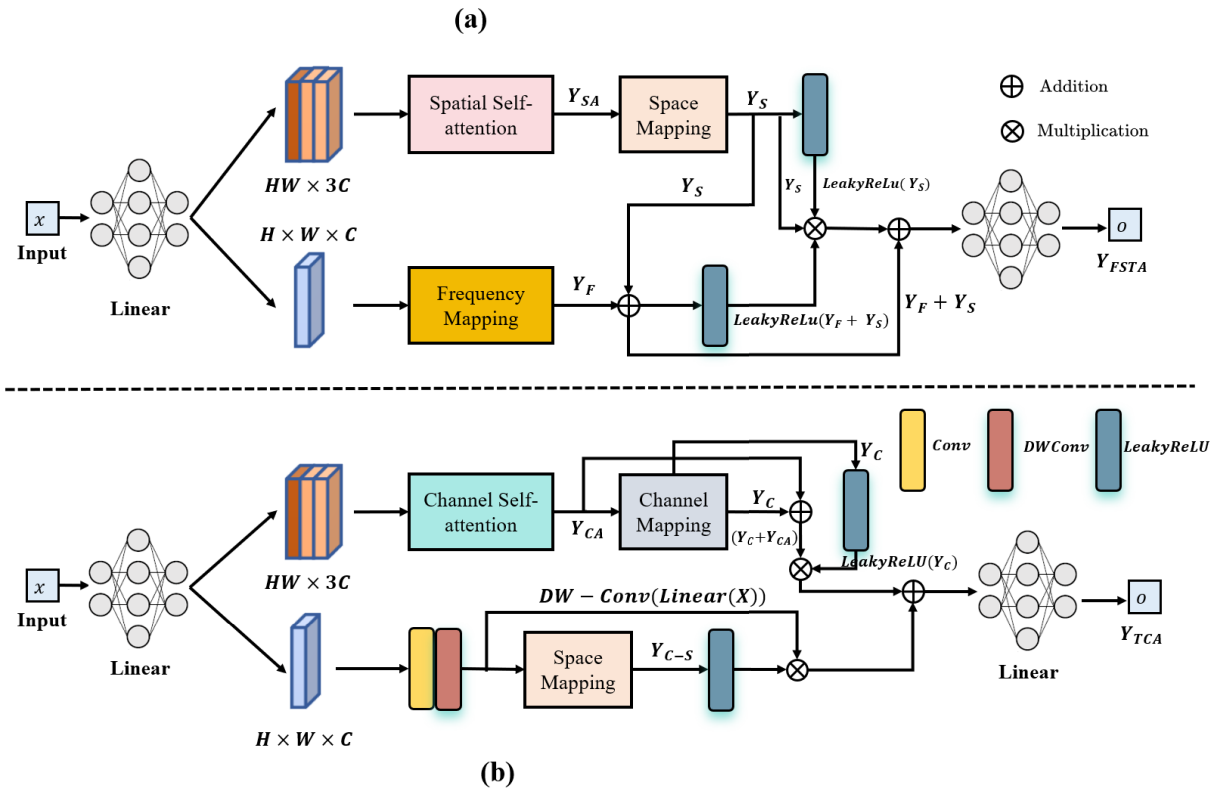


Fig. 4. Schematic illustrations of the FSTA and TCA network structures. (a) Framework of FSTA. (a) Framework of TCA.

The visualization result of the amplitude obtained by taking the absolute value of the above high-frequency components with high frequency (absolute) has the core function of eliminating the positive and negative signal differences caused by phase polarity and only retaining the intensity distribution characteristics of high-frequency information (such as the signal amplitude size at the edge, the density of texture, and the contrast strength of local details). Compared with the original high-frequency components, this graph can more intuitively highlight the spatial positions of high-frequency details. This indicates that the splitting of FDB is not random but can specifically capture the key frequency components in the input that carry the recognition of details (such as potential fine edges and texture regions in the image).

B. Parallel Frequency-Aware Transformer Module

As existing studies [14] have pointed out, the self-attention mechanism exhibits low-pass filtering characteristics and, thus, often has difficulty effectively restoring high-frequency details in images. To this end, we propose a PFTM to enhance the reconstruction capability of high-frequency details. Specifically, the PFTM structure (as shown in Fig. 2) consists of two parallel attention units: frequency–spatial transposed attention (FSTA) and transposed channel attention (TCA).

1) Frequency–Spatial Transposed Attention: To further enhance the concentration of high-frequency components, we constructed FSTA, whose structure is shown in Fig. 4. This module combines high-frequency information with spatial and frequency features and embeds them into the self-attention calculation process, thereby effectively improving the reconstruction quality of high-frequency details.

Specifically, this method first divides the input features into multiple nonoverlapping rectangular windows of size $(H \times W)/(wh \times ww)$. For the feature $X^i \in \mathbb{R}^{(wh \times ww) \times C}$ window, we generate the query (Q), key (K), and value (V) vectors, respectively, through unbiased linear projection, denoted as Q^i_{FSTA}, K^i_{FSTA}, and V^i_{FSTA}. The calculation of self-attention can be expressed as

$$Y^i_{\text{SA}} = \text{SoftMax}\left(\frac{Q^i_{\text{FSTA}}\left(K^i_{\text{FSTA}}\right)^T}{\sqrt{d}} + B\right) \cdot V^i_{\text{FSTA}} \tag{4}$$

where B denotes the relative position bias. Furthermore, adopting the approach introduced in Swin transformer [41], we incorporate a shifted window partitioning strategy to strengthen the model's capacity for capturing global spatial context. Subsequently, the output features Y^i_{SA} from all windows are reshaped and concatenated, yielding the final spatial self-attention output Y_{SA}.

To extract spatial and frequency features, we employ a multibranch approach using Y_{SA} and the input X, incorporating both space mapping (SM) and frequency mapping modules (FMs). These modules perform high-dimensional mapping and transformation on the features along the spatial and frequency dimensions, respectively,

$$Y_S = f_{\text{SM}}\left(Y_{\text{SA}}\right) \tag{5}$$

$$Y_F = f_{\text{FM}}\left(\text{Linear}\left(X\right)\right) \tag{6}$$

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where Y_S denotes the output after spatial transformation. Specifically, we employ a combination of 1×1 convolution and depthwise convolution to extract spatial information from Y_{SA}, followed by channel splitting. To model more complex spatial feature representations, nonlinear transformations are introduced over half of the channels. The spatial mapping process is finalized by aggregating features along the channel dimension.

On the other hand, Y_F corresponds to the extraction and transformation of frequency information. This component begins by projecting the input X into the frequency feature space via a fully connected layer. Global high-frequency information is captured using a 1×1 convolution, followed by a 3×3 max-pooling operation. Subsequently, a series of 1×1 convolutions and activation layers are applied to integrate both global and local high-frequency information, resulting in an enhanced feature representation Y_F with emphasized high-frequency components.

Finally, feature fusion is accomplished via cross-branch weighting. The procedure is given as follows:

$$Y_{\mathrm{FSTA}} = (Y_S \oplus Y_F) + \left(Y_S \odot \mathrm{LeakyReLU}\left(Y_S\right)\right) \odot \left(Y_S \odot \mathrm{LeakyReLU}\left(Y_S + Y_F\right)\right) \tag{7}$$

where LeakyReLU$(\cdot)$ denotes the LeakyReLU activation function. The cross-branch weighting strategy enables the integration of frequency information often absent in self-attention, along with globally attended low-frequency components.

2) Transposed Channel Attention: TCA employs a distinct approach from FSTA by performing self-attention across the channel dimension. Similar to common practice, channels are divided into multiple heads, and channel attention is applied, as depicted in Fig. 4. For the input feature $X \in \mathbb{R}^{B \times H \times W \times C}$, the queries, keys, and values (denoted as Q_{CTA}, K_{CTA}, and K_{CTA}) in the self-attention mechanism are computed through the following steps:

$$Y_{\mathrm{CA}} = \mathrm{SoftMax}\left(\frac{Q_{\mathrm{CTA}}^T\left(K_{\mathrm{CTA}}\right)}{\alpha}\right) \cdot V_{\mathrm{CTA}} \tag{8}$$

where α is a learnable temperature parameter that scales the dot product. Finally, the channel-aware feature Y_{CA} is obtained by reshaping the concatenated outputs from multiple attention heads. Subsequently, the channel mapping (CM) is employed to extract channel features. Specifically, a sequence of 1×1 and depthwise convolutional operations is performed on $X \in \mathbb{R}^{B \times H \times W \times C}$ to integrate both global and local information, yielding the initial channel representation Y_C and Y_{C-S}

$$Y_C = f_{\mathrm{CM}}\left(Y_{\mathrm{CA}}\right) \tag{9}$$

$$Y_{C-S} = f_{\mathrm{SM}}\left(f_{\mathrm{DW-Conv}}\left(\mathrm{Linear}\left(X\right)\right)\right) \tag{10}$$

where f_{CM} represents the channel mapping, Y_{C-S} represents the spatial mapping feature in TCA, f_{SM} represents the spatial projection, and $f_{\mathrm{DW-Conv}}$ is 1×1 and widthwise convolutional operations. Then, information extraction and cross-weight transfer are carried out as follows:

$$Y_{\mathrm{TCA}} = \left[\left(f_{\mathrm{DW-Conv}}\left(\mathrm{Linear}\left(X\right)\right)\right) \odot \mathrm{LeakyReLU}\left(Y_{C-S}\right)\right] \oplus \left[\left(Y_C \oplus Y_{\mathrm{CA}}\right) \odot \mathrm{LeakyReLU}\left(Y_C\right)\right] \tag{11}$$

where $\oplus$ is the elementwise addition and $\odot$ is the elementwise multiplication. This strategy merges information from complementary spatial and channel sources.

3) Frequency Dual-Aggregation Network: To effectively integrate the strengths of transformers and CNNs for frequency dual aggregation, we propose a frequency dual-aggregation network (FDAN), as illustrated in Fig. 2. Specifically, the input feature (Y_{FSTA} or Y_{TCA}) is first projected via a fully connected layer onto an intermediate representation X_{in}, followed by a GELU activation. A frequency gate f_g is then applied to extract frequency-domain information. This gate consists of two convolutional layers that decompose the input features into low- and high-frequency components. The low-frequency component remains unchanged, while the high-frequency component is enhanced through a series of 1×1 convolutions and depthwise convolutional (DW-Conv) layers

$$f_g = X_{\mathrm{in}} \cdot f_{\mathrm{DW-Conv}}\left(\mathrm{Conv}_{1\times1}\left(X_{\mathrm{in}}\right)\right) \tag{12}$$

$$X_{\mathrm{in}} = f_{\mathrm{GELU}}\left(\mathrm{Linear}\left(x\right)\right) \tag{13}$$

where f_g is obtained by elementwise multiplication of the two components, producing a feature map that integrates both low- and high-frequency information. Finally, another fully connected layer projects the fused features back to the original feature space as Y_{FDAN}. With this design, the DFFN effectively extracts frequency-domain information while preserving spatial-domain characteristics.

At the end of FDAN, we combined a series of 1×1 convolution and depthwise convolution based on depth at the end of each PFTM group. This was to compensate for the loss of detailed information caused by long-distance jump connections, thereby enhancing the recovery of high-frequency details.

It should be emphasized that FDB and FDAN perform operations of different granularities and different target frequencies. FDB forcibly decomposes the input image at the macrolevel to generate clearly separated low- and high-frequency branches for parallel processing. FDAN, on the other hand, operates at the micro level of feature fusion as a soft and adaptive frequency weight adjustment mechanism. It dynamically learns and emphasizes the frequency components that most need to be enhanced in the current context (whether from low-frequency or high-frequency branches), thereby achieving more refined feature regulation. Therefore, this is not a repetitive operation but rather an organic combination of "coarse separation" and "fine enhancement" under the "divide and conquer" strategy.

C. Loss Function

Following established practice, the L_1 loss is employed to minimize the distance between the SR image I_{SR} and the ground-truth HR image I_{HR}, which is formulated as

$$\mathcal{L}_{\mathrm{SR}} = \left\|I_{\mathrm{SR}} - I_{\mathrm{HR}}\right\|_1. \tag{14}$$

IV. Experiment Results and Discussion

A. Dataset

In this study, we employ two widely used remote sensing image datasets: UCMerced [42] and PatternNet [43].

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The UCMerced dataset, developed by the UC Merced Computer Vision Laboratory, is designed for the classification of remote sensing image scenes. It consists of images manually cropped from large-scale urban area imagery in the USGS National Map. The dataset encompasses 21 scene categories, each containing 100 images of size 256 × 256 pixels. We use 90 images per category for training and validation, and the remaining ten for testing, resulting in 1890 training/validation images and 210 test images.

PatternNet, a large-scale HR remote sensing dataset, is designed for image retrieval tasks. It includes 38 classes, each with 800 images of size 256 × 256 pixels, sourced from Google Earth imagery and the Google Maps API. For our experiments, 62 images per class are allocated for training and validation, and eight images per class are reserved for testing, yielding a total of 2356 training/validation images and 304 test images.

B. Implementation Details

The proposed model is trained using the Adam optimizer [44] with coefficients $\beta_1 = 0.9$, $\beta_2 = 0.99$, and $\varepsilon = 10^{-8}$. The input consists of RGB images, augmented via random horizontal and vertical flipping during training. Each training batch contains 32 randomly cropped LR patches of size 64 × 64. The model is trained for 500 000 backpropagation iterations in total. The initial learning rate is set to 5×10^{-4} and is reduced by half every 125 000 iterations.

All experiments are implemented in PyTorch and executed on an NVIDIA GeForce GTX 4090 GPU. The training and testing images are downsampled by factors of ×2 and ×4 to produce LR images with resolutions of 128 × 128 and 64 × 64, respectively. Quantitative evaluation is conducted using PSNR, structural similarity index measure (SSIM) [45], and feature similarity index measure (FSIM) [46]. A summary of all the datasets, methods, and evaluation metrics is shown in Table I.

C. Comparison With the SOTA

In this section, we compare the proposed model with several SOTA SR methods, including HAN [19], MHAN [8], CTNet [9], HSENet [6], HAUNet [12], LKFN [15], and MSGAN [27]. Among these, HAN and LKFN are established SR approaches designed for general images, while MHAN, CTNet, HSENet, HAUNet, and MSGAN represent recent SOTA methods specifically developed for remote sensing images. To ensure a fair comparison, all models were trained and tested under consistent data conditions using the authors' official implementations.

1) Quantitative Comparison: Tables II and III summarize the SR reconstruction outcomes on two public datasets. The best results are highlighted in red in each row. Across all three test sets, FDFNet achieves the highest performance in terms of PSNR, SSIM, and FSIM. On the UCMerced test set, with a scale factor of 2, FDFNet exceeds the second-best method, HAUNet and LKFN, by an average of 0.49 dB in PSNR; at a scale factor of 4, the margin increases to 0.25 dB. Similarly, on the PatternNet dataset, FDFNet outperforms MHAN by 0.32 and 0.28 dB for scale factors of 2 and 4, respectively.

2) Qualitative Comparison: Figs. 5 and 6, respectively, present the visual comparison results of UCMerced and PatternNet under the ×4 scale factor. It can be observed from the visualization effect that FDFNet can restore structures with rich textures and clear edges, especially well preserving the high-frequency details in remote sensing images. Taking the reconstructed stadium and plot division lines in Fig. 5 as an example, the recently proposed competition methods based on transformer (such as HAUNet and LKFN) all have difficulty accurately reconstructing their contour and texture information. In terms of vehicle reconstruction, the proposed method is the closest to the real HR image in high-frequency detail restoration. It can clearly present structures such as windshields. However, other methods generally result in structural distortion and texture distortion. These visual contrasts further verify the effectiveness of FDFNet in capturing and reconstructing fine structures.

When comparing the visual results of the top images in the PatternNet dataset in Fig. 6, we found that all SR networks based on CNNs and transformers struggled to handle this issue, resulting in suboptimal results and the loss of some high-frequency textures. In contrast, thanks to the decoupling and divide-and-control capability of FDB, as well as the triple interaction exploration of the spatial frequency channel, the results generated by our FDFNet visually approach the ground truth (GT) and successfully generate multifrequency lines. For instance, in the reconstruction of coils and lane lines on a basketball court, all other methods produced significant blurring. In contrast, our method still maintained good visual quality and more high-frequency contextual information.

Combined with Table IV, in the SR task with an input resolution of 64 × 64 and the UCMerced test set (×4 scale factor), the proposed model has a relatively low parameter count of 0.89M and achieves the optimal performance in both inference efficiency and SR quality, with the shortest inference time of 0.04 s and the highest PSNR of 27.33 dB. CTNet has a mere parameter count of 0.45M and a fast inference time of 0.05 s, yet its PSNR of 26.54 dB is relatively low. LFKN possesses the smallest parameter count of 0.31M among all models, an inference time of 0.11 s, and a PSNR of 27.08 dB, whose SR performance is only second to the proposed method. In contrast, HAN has a considerably high parameter count of 15.42M, and MHAN has 11.35M parameters along with a long inference time of 0.65 s; both models not only suffer from high complexity and poor inference efficiency but also have lower PSNR values (26.84 and 26.69 dB, respectively) than the proposed method. To sum up, the proposed method achieves a favorable balance among parameter count, inference efficiency, and SR performance, while other models either excel in lightweight design and efficiency but lack in performance or feature large parameter counts, low efficiency, and mediocre performance.

D. Ablation Study

A comprehensive evaluation of the proposed FDFNet is provided in this section through an ablation study, which examines the impact of its core architectural components and alternative configurations. For consistency, every model variant

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TABLE I
SUMMARY OF DATASETS AND EVALUATION METRICS FOR SOTA IMAGE SR METHODS

Method	Year/Venue	Training Dataset	Testing Dataset	Evaluation Metrics	Key Notes
HAN [19]	2020 / ECCV	UCMerced(1890 for train and val) [42], PatternNet(2356 for train and val) [43]	UCMerced(210 for test) [42], PatternNet(304 for test) [43]	Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index Measure (SSIM) [45], Feature Similarity Index Measure (FSIM) [46]	Proposed a Holistic Attention Network (HAN) with LAM and CSAM to model layer-channel-position interdependencies for SISR.
MHAN [8]	2021 / TGRS	UCMerced(1890 for train and val) [42], PatternNet(2356 for train and val) [43]	UCMerced(210 for test) [42], PatternNet(304 for test) [43]	Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index Measure (SSIM) [45], Feature Similarity Index Measure (FSIM) [46]	Proposed MHAN for remote sensing SR: weighted channelwise concatenation in skip connections, HOA refinement, and frequency-aware connections.
CTNet [9]	2022 / TGRS	UCMerced(1890 for train and val) [42], PatternNet(2356 for train and val) [43]	UCMerced(210 for test) [42], PatternNet(304 for test) [43]	Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index Measure (SSIM) [45], Feature Similarity Index Measure (FSIM) [46]	Proposed CTL (lightweight convolution replacing 3×3 convolutions) and built CTN with CTB, CEM for lightweight remote-sensing image super-resolution.
HSENet [6]	2022 / TGRS	UCMerced(1890 for train and val) [42], PatternNet(2356 for train and val) [43]	UCMerced(210 for test) [42], PatternNet(304 for test) [43]	Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index Measure (SSIM) [45], Feature Similarity Index Measure (FSIM) [46]	Proposed HSENet for remote sensing image SR, using HSEM (integrating SSEM for single-scale and CCS for cross-scale self-similarity) to exploit hybrid-scale similarities.
HAUNet [12]	2023 / TGRS	UCMerced(1890 for train and val) [42], PatternNet(2356 for train and val) [43]	UCMerced(210 for test) [42], PatternNet(304 for test) [43]	Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index Measure (SSIM) [45], Feature Similarity Index Measure (FSIM) [46]	Proposed HAUNet (with two conv attention SEMs and CIM) and lightweight HAUNet-S for RSISR, using hybrid convolution attention to explore multiscale features and enhance global representation, boosting performance.
LKFN [15]	2024 / CVPR	UCMerced(1890 for train and val) [42], PatternNet(2356 for train and val) [43]	UCMerced(210 for test) [42], PatternNet(304 for test) [43]	Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index Measure (SSIM) [45], Feature Similarity Index Measure (FSIM) [46]	Proposed LKFN for efficient super-resolution, adopting FPA (frequency-domain attention), large kernel/partial convolutions to capture long-range dependencies, achieving SOTA with lightweight design.
MSGAN [27]	2025 / PR	UCMerced(1890 for train and val) [42], PatternNet(2356 for train and val) [43]	UCMerced(210 for test) [42], PatternNet(304 for test) [43]	Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index Measure (SSIM) [45], Feature Similarity Index Measure (FSIM) [46]	Proposed a streamlined RS-SR framework via MSGAN, with a novel multi-scale graph attention module (integrating graph attention and multi-scale lattice blocks) to balance computational efficiency and performance.

was trained under identical conditions on the UCMerced dataset using a magnification factor of $\times2$.

1) Effectiveness of the FSTA, TCA, FDB, and FDAN: To evaluate the efficacy of the FSTA, TCA, FDB, and FDAN, we performed a series of ablation experiments. As listed in the first row of Table V, after removing FDB, the model's PSNR dropped to 33.84 dB. The core function of FDB is to decouple the frequency information (high-frequency details and low-frequency structure) in the input features. When FDB is missing, high- and low-frequency features are superimposed, making it difficult for subsequent modules to handle different frequency components specifically, which directly leads to a decline in reconstruction accuracy. After removing TCA, the PSNR was only 33.81 dB. Through the channel attention under the transpose mechanism, TCA dynamically assigns weights to different channels, which can specifically enhance the channel features corresponding to key ground objects (such as building contours and vegetation textures) in remote sensing images. When TCA is missing, the effective information between channels is submerged by redundant information,

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Fig. 5. Visual comparison of different methods on the UCMerced 4× test set.

TABLE II

AVERAGE PERFORMANCE METRICS ON THE UCMERCED [42] TEST SET. THE BEST RESULTS ARE HIGHLIGHTED IN RED, AND THE SECOND-BEST ARE MARKED IN BLUE

Method	Scale	PSNR/dB	SSIM [45]	FSIM [46]
Bicubic	×2	29.73	0.8689	0.9083
HAN [19]		33.32	0.9313	0.9605
MHAN [8]		33.05	0.9288	0.9583
CTNet [9]		32.57	0.9235	0.9536
HSENet [6]		33.37	0.9329	0.9618
HAUNet [12]		33.43	0.9333	0.9618
LKFN [15]		33.40	0.9328	0.9614
MSGAN [27]		32.81	0.9255	0.9558
Ours		33.92	0.9389	0.9659
Bicubic	×4	25.05	0.6624	0.7600
HAN [19]		26.84	0.7543	0.8378
MHAN [8]		26.69	0.7471	0.8311
CTNet [9]		26.54	0.7390	0.8220
HSENet [6]		26.98	0.7562	0.8404
HAUNet [12]		27.04	0.7613	0.8422
LKFN [15]		27.08	0.7637	0.8425
MSGAN [27]		26.67	0.7424	0.8269
Ours		27.33	0.7722	0.8537

TABLE III

AVERAGE PERFORMANCE METRICS ON THE [43] TEST SET. THE BEST RESULTS ARE HIGHLIGHTED IN RED, AND THE SECOND-BEST ARE MARKED IN BLUE

Method	Scale	PSNR/dB	SSIM [45]	FSIM [46]
Bicubic	×2	32.43	0.8952	0.9257
HAN [19]		35.84	0.9377	0.9658
MHAN [8]		35.86	0.9375	0.9655
CTNet [9]		35.36	0.9339	0.9628
HSENet [6]		35.73	0.9372	0.9656
HAUNet [12]		35.60	0.9350	0.9631
LKFN [15]		35.56	0.9345	0.9635
MSGAN [27]		35.62	0.9355	0.9639
Ours		36.18	0.9358	0.9669
Bicubic	×4	26.66	0.7081	0.7889
HAN [19]		29.18	0.7966	0.8631
MHAN [8]		28.89	0.7919	0.8567
CTNet [9]		28.90	0.7888	0.8506
HSENet [6]		29.34	0.8012	0.8641
HAUNet [12]		28.91	0.7906	0.8551
LKFN [15]		29.37	0.8006	0.8619
MSGAN [27]		28.94	0.7902	0.8563
Ours		29.65	0.8072	0.8711

which restricts the reconstruction performance. After removing FSTA, the PSNR dropped to 33.88 dB. The role of FSTA is to achieve cross-domain interaction between frequency and spatial dimension features: on the one hand, it optimizes the fineness of the spatial structure by using frequency information; on the other hand, it constrains the rationality of

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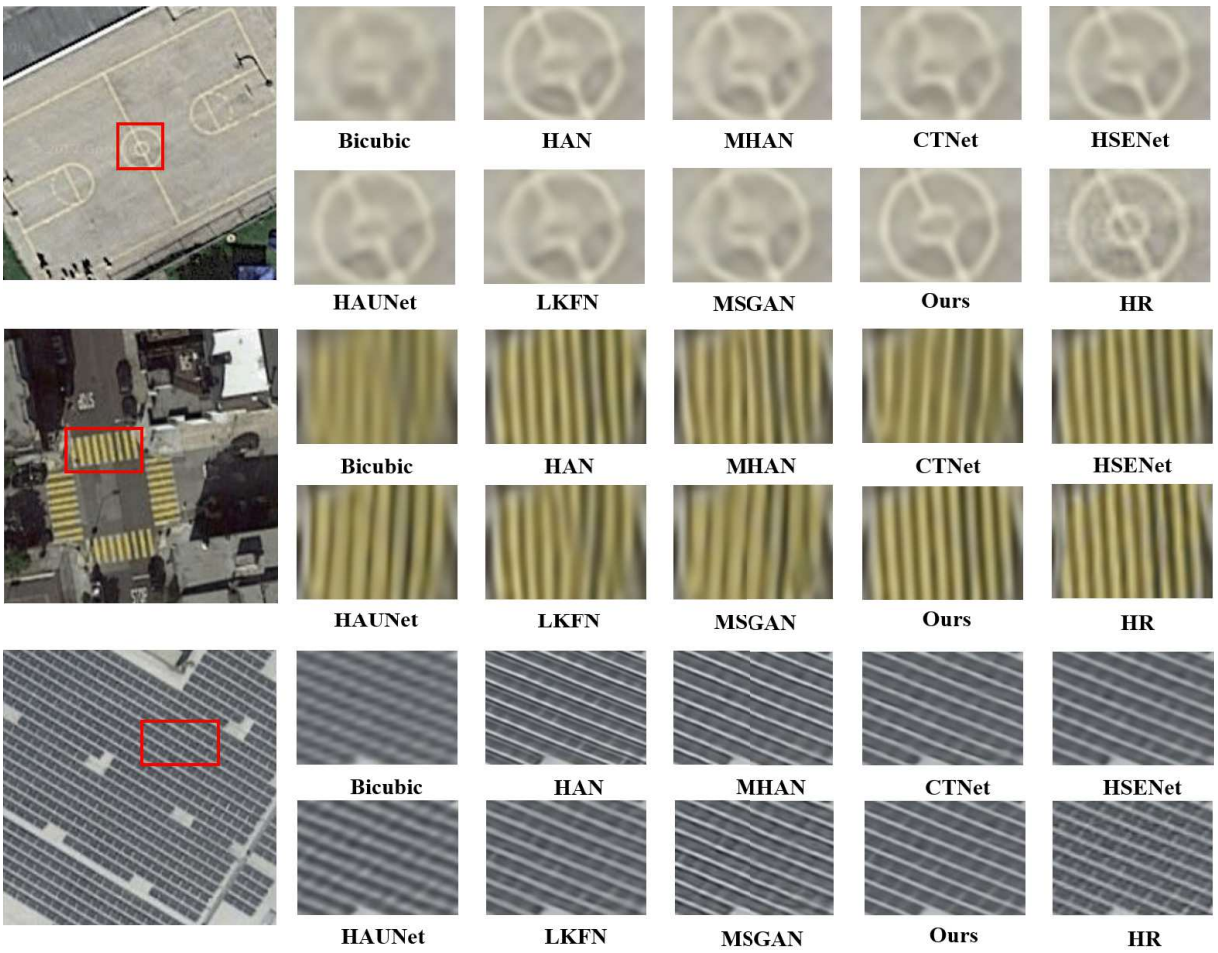


Fig. 6. Visual comparison of different methods on the PatternNet 4× test set.

TABLE IV

COMPARISON OF INFERENCE TIME AND PARAMETER QUANTITY OF DIFFERENT MODELS: THE TEST INPUT RESOLUTION IS 64 × 64, AND THE TEST SET IS UCMERCED [42] WITH SCALE ×4. THE BEST RESULTS ARE HIGHLIGHTED IN RED, AND THE SECOND-BEST ARE MARKED IN BLUE

Methods	Params(M) ↓	Inference time(s) ↓	PSNR ↑	SSIM ↑
HAN [19]	15.42	0.11	26.84	0.7543
MHAN [8]	11.35	0.65	26.69	0.7471
CTNet [9]	0.45	0.05	26.54	0.7390
HSENet [6]	5.43	0.34	26.98	0.7562
HAUNet [12]	2.66	0.05	27.04	0.7613
LKFN [15]	0.31	0.11	27.08	0.7637
MSGAN [27]	1.62	0.29	26.67	0.7424
Ours	0.89	0.04	27.33	0.7722

TABLE V

ABLATION STUDY ON THE IMPACT OF FSTA, TCA, FDB, AND FDAN ON RECONSTRUCTION PERFORMANCE. "W/O" DENOTES THE REMOVAL OF A SPECIFIC COMPONENT. EXPERIMENTS ARE CONDUCTED ON THE UCMERCED TEST SET AT A SCALE FACTOR OF ×2. THE BEST RESULTS ARE HIGHLIGHT

Model settings	PSNR/dB	SSIM [45]	FSIM [46]
FDFNet w/o FDB	33.84	0.9365	0.9648
FDFNet w/o TCA	33.81	0.9367	0.9651
FDFNet w/o FSTA	33.88	0.9379	0.9655
FDFNet w/o FDAN	33.71	0.9366	0.9648
FDFNet (the finals)	**33.92**	**0.9389**	**0.9659**

frequency details through the spatial structure. When FSTA is missing, the matching degree between the details (frequency) and the structure (space) of the reconstructed image decreases, and thus, the accuracy is reduced. After removing FDAN, the PSNR is 33.71 dB, and FDAN is responsible for the dynamic aggregation of features. Since FDB, TCA, and FSTA have completed the work of "frequency decoupling–channel regulation–cross-domain interaction," the performance loss after the absence is the largest, reflecting its optimization value for feature fusion.

Fig. 7 shows the residual heat maps of different models. The residual brightness is positively correlated with the reconstruction error (the higher the brightness, the greater the difference between the SR image and the HR image, and the lower the reconstruction accuracy). When FDB is missing, the bright colors in the "high-frequency detail density areas" (such as the texture of building roofs and the edges of road lane lines) in the heat map are more obvious. The absence of "frequency decomposition decoupling" in FDB leads to the aliasing of high-frequency details and low-frequency structures, making it difficult to accurately reconstruct fine textures, and the

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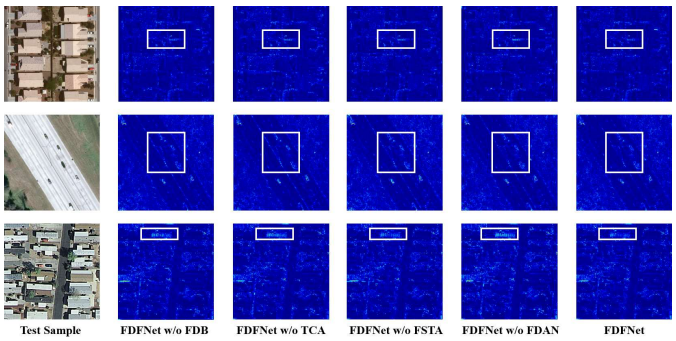


Fig. 7. Residual heat map comparison of FDB/TCA/FSTA/FDAN ablation (UCMerced dataset, 2× scaling).

TABLE VI

ABLATION STUDY ON THE IMPACT OF SM, FM, AND CM ON RECONSTRUCTION PERFORMANCE. "W/O" DENOTES THE REMOVAL OF A SPECIFIC COMPONENT. EXPERIMENTS ARE CONDUCTED ON THE UCMERCED TEST SET AT A SCALE FACTOR OF ×2. THE BEST RESULTS ARE HIGHLIGHT

Model settings	PSNR/dB	SSIM [45]	FSIM [46]
FDFNet w/o SM	33.82	0.9375	0.9652
FDFNet w/o FM	33.86	0.9369	0.9655
FDFNet w/o CM	33.85	0.9378	0.9649
FDFNet (the finals)	**33.92**	**0.9389**	**0.9659**

residuals are concentrated in the detail areas. When TCA is removed, the bright color residuals are more prominent in "feature boundaries and category transition zones" (such as the junctions of buildings and vegetation, and the boundaries of roads and surrounding land). This reflects the lack of "precise regulation of channel features" by TCA, which leads to a decline in the discrimination of channel features among different features and an increase in the reconstruction errors of feature boundaries and category features. When FSTA is missing, the bright color residuals are denser in areas with high "frequency–spatial feature interaction requirements" (such as the spatial layout of complex building textures and the junction of road direction and pavement texture). The absence of "frequency–space cross-domain interaction" in FSTA was verified, which led to a decrease in the matching degree between frequency details and spatial structure, and insufficient feature synergy caused residuals. When the FDAN deficiency is removed, the overall bright colors in the heat map are the most prominent, and the range of bright colors is the widest. It reflects the lack of dynamic feature aggregation in FDAN, which makes the features processed by the previous module unable to be efficiently coordinated, and the accumulation of errors leads to the maximization of residuals. The final complete heat map of FDFNet is the darkest overall, with only a very small number of "ultrahigh-frequency detail limit regions" showing faint bright colors.

2) Effectiveness of the Mapping Strategy: The validity of the projection component is evaluated in Table VI. After removing SM, the model's PSNR dropped to 33.82 dB. SM extracts spatial features, introduces nonlinear transformations to half of the channels to model complex spatial structures,

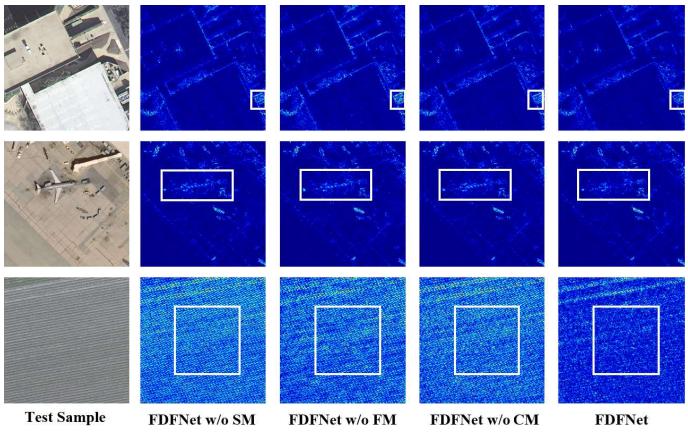


Fig. 8. Residual heat map comparison of SM/FM/CM ablation (UCMerced dataset, 2× scaling).

and ultimately enhances spatial detail representation through channel dimension aggregation. This result indicates that SM can effectively capture structured spatial information such as building contours and road boundaries in remote sensing images. The absence of such information will lead to the blurring and misalignment of spatial textures, directly affecting the reconstruction accuracy. After removing FM, the PSNR dropped to 33.86 dB. The core function of FM is to project input features into the frequency space, thereby capturing global high-frequency information. Subsequently, it integrates global–local high-frequency features through Max pooling and the activation layer, with a focus on enhancing the recovery of high-frequency details such as edges and textures. The performance degradation verified the FM's precise ability to capture high-frequency information. After removing the CM, the PSNR dropped to 33.85 dB. CM integrates the global and local information of channel dimensions, solving the limitation of "restoring features only by relying on spatial dimensions." The performance degradation indicates that CM can enhance the feature interaction between channels, enabling the model to more accurately utilize cross-channel information to optimize feature representation. This is particularly significant for distinguishing the features of different ground objects (such as vegetation and buildings) in remote sensing images.

Fig. 8 shows the residual heat maps of different models. The residual brightness is positively correlated with the reconstruction error (the higher the brightness, the greater the difference between the SR image and the HR image, and the lower the reconstruction accuracy). For the complete FDFNet, the residual heat map is the darkest overall, with only a few high-frequency regions (such as building edges and texture details) showing weak brightness, indicating that the model has the smallest reconstruction error for spatial structure and high-frequency details, and the highest consistency with HR images. After the removal of SM, the residual map has the highest brightness, especially in areas with complex spatial structures (such as dense buildings and road intersections), where obvious bright spots are present. This corresponds to the largest PSNR decrease in quantitative indicators, verifying the inhibitory effect of SM on spatial detail errors. When FM is removed, the residual map shows significantly higher

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TABLE VII

AVERAGE EVALUATION METRICS ON THE OPENBAYES TEST SET (SCALE FACTOR ×4). THE BEST AND SUBOPTIMAL RESULTS ARE HIGHLIGHTED IN RED AND BLUE, RESPECTIVELY

Method	PSNR/dB	SSIM [45]	FSIM [46]
Bicubic	25.54	0.6860	0.9021
HAN [19]	26.86	0.7606	0.9461
MHAN [8]	26.95	0.7619	0.9454
CTNet [9]	26.95	0.7582	0.9434
HSENet [6]	27.13	0.7673	0.9488
HAUNet [12]	27.07	0.7659	0.9484
LKFN [15]	27.01	0.7625	0.9481
MSGAN [27]	26.98	0.7610	0.9460
FDFNet(Ours)	27.27	0.7736	0.9508

TABLE VIII

AVERAGE METRICS DERIVED BY THE DEEPLABV3+ SEGMENTATION METHOD. THE BEST RESULTS ARE HIGHLIGHT

Method	Overall Acc	Mean Acc	Mean IoU
Bicubic+DeepLabV3+	0.386379	0.328649	0.171311
HAN+DeepLabV3+	0.392355	0.315961	0.143003
MHAN+DeepLabV3+	0.383429	0.311065	0.141949
CTNet+DeepLabV3+	0.389627	0.323166	0.142822
HSENet+DeepLabV3+	0.378254	0.303319	0.140492
HAUNet+DeepLabV3+	0.397862	0.318158	0.147727
LKFN+DeepLabV3+	0.371193	0.292899	0.136738
MSGAN+DeepLabV3+	0.394411	0.322642	0.146333
FDFNet+DeepLabV3+(Ours)	**0.408671**	**0.329876**	**0.189904**

brightness in high-frequency texture regions (such as vegetation textures and railing details) than the complete model, indicating that the reconstruction error of high-frequency details increases after FM is missing, resulting in blurred image textures. After removing the CM, the residual map has a higher brightness at the boundaries of different features (such as the junction of vegetation and buildings), reflecting the decrease in the discrimination of feature characteristics caused by insufficient integration of channel information.

This indicates that SM, FM, and CM, respectively, construct feature representation capabilities from three dimensions: spatial structure, frequency details, and channel interaction, forming a 3-D collaborative mechanism of "space–frequency–channel": SM lays the foundation of spatial structure, FM supplements high-frequency detail information, and CM enhances cross-channel feature fusion. The three jointly support FDFNet's precise reconstruction of complex scenes in remote sensing images.

E. Image Segmentation Verification SR

To evaluate the generalization capability of the proposed model, SR reconstruction and semantic segmentation experiments were carried out on the OpenBayes test set. This dataset comprises 303 remote sensing images acquired from Google Earth, covering an urban area in southern China. Each image has a size of 560 × 560 pixels and a spatial resolution of 1.2 m. For the segmentation task, DeepLabv3+ [47] was employed as the baseline model. This segmentation model was pretrained on large-scale datasets (such as ImageNet), and we only fine-tuned it on the OpenBayes dataset. This transfer learning strategy requires much less data volume than ab initio training. The first 273 images were used for training and validation, and the remaining 30 images were reserved for testing. In the SR task, the model pretrained on the UCMerced dataset was applied directly without any fine-tuning or retraining. Quantitative results of the SR reconstruction are presented in Table VII. As shown in the table, the proposed model achieves superior performance compared to other methods, demonstrating its enhanced generalization ability over SOTA SR approaches.

Furthermore, the effectiveness of the proposed model was evaluated through high-level vision tasks, including semantic segmentation. For the 30 test images, SR reconstruction was first performed using several SOTA models, and the resulting images were subsequently compared in terms of segmentation accuracy. The evaluation employed widely adopted metrics in semantic segmentation: mean accuracy (mAcc), overall accuracy (OA), and mean Intersection over Union (mIoU). Detailed results are provided in Table VIII. The proposed model achieved an mIoU of 0.189904, exceeding the second-best method by 10.8%, and an OA of 0.408671, which represents a 2.7% improvement over the next highest result, where the mAcc value is higher than the next highest by 2.2%, demonstrating its comprehensive effectiveness. Visual comparisons of the segmentation results are presented in Fig. 9.

As shown in Fig. 9, the proposed method produces segmentation results that align more closely with the GT, accurately delineating finer satellite objects with significantly reduced misclassification and omission. These outcomes indicate superior segmentation performance. In contrast, other SR methods exhibit limitations in edge delineation and introduce noticeable noise and multiclass recognition errors. Since semantic segmentation is a pixel-level prediction task, the structural and textual information of the image considerably influences the performance. The improved segmentation results achieved using our SR reconstructions confirm the effectiveness of the proposed method.

F. Limitations and Future Work

Limitations of the Model: The current FDB module relies on the fixed-parameter DTCWT to achieve frequency separation. Although it can effectively handle the low/high-frequency components of conventional remote sensing images, its adaptability is limited when facing extreme scenarios. For instance, in strongly noisy remote sensing images (such as those with cloud and fog interference), noise signals are prone to being misjudged as high-frequency details, resulting in the amplification of noise in the reconstruction results. However, in areas with dense ultrasmall targets (such as dense building complexes and small vegetation patches), a fixed frequency division threshold may lead to the loss of high-frequency features of some ultrasmall targets, affecting the accuracy of detail reconstruction.

The existing models perform well at ×2 and ×4 scales (for example, the PSNR of the UCMerced dataset at ×4 scale reaches 27.33 dB, and that of the PatternNet dataset at

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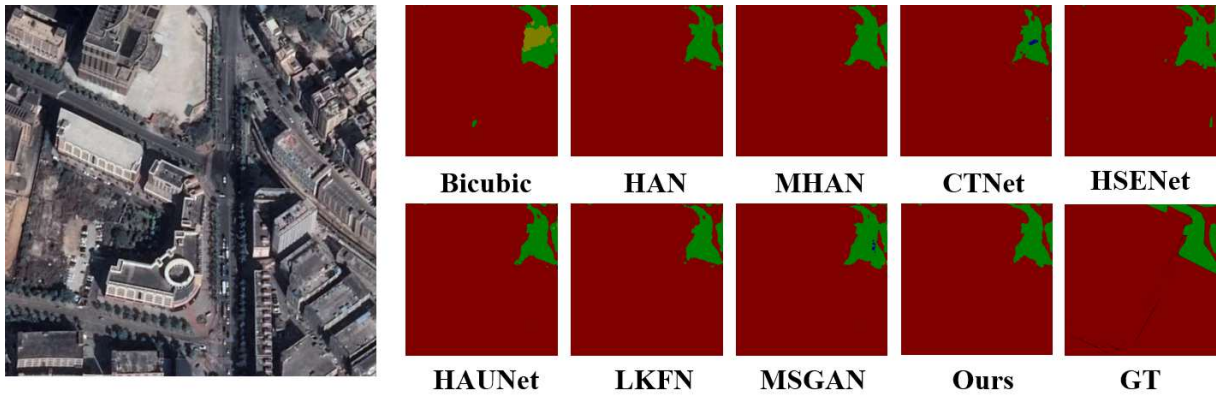


Fig. 9. Visualization of the results obtained by the DeepLabV3+ segmentation method.

$\times 4$ scale reaches 29.65 dB), but, when it comes to the SR task of remote sensing images at $\times 8$ and larger scales, there is a significant decline in performance. This is because the attenuation of high-frequency details is more significant on a large scale. The dual-attention mechanism of the existing PFTM module is difficult to fully capture the high-frequency correlation information across scales, and information loss is prone to occur during the feature transfer process, resulting in insufficient fine texture detail recovery.

Future Work: We plan to carry out subsequent research work in three aspects. First, we plan to introduce a dynamic frequency division strategy based on image content. By designing a lightweight content perception module, we can analyze the noise intensity, target scale distribution, and other features of the input image in real time and dynamically adjust the frequency division parameters of DTCWT. Meanwhile, a new noise-detail distinction branch is added. The attention mechanism is utilized to accurately identify and separate the noise and effective details in the high-frequency components, enhancing the model's adaptability to complex scenarios.

Second, for large-scale SR tasks, a cross-scale feature interaction module is designed to construct a bidirectional fusion channel of shallow high-frequency features and deep semantic features, and information loss is reduced through multistage feature distillation technology. Meanwhile, the frequency aggregation strategy of the FDAN module is optimized, and scale adaptive weights are introduced to enable the model to dynamically adjust the aggregation intensity of frequency information according to the SR scale, maintaining excellent detail recovery capability even at $\times 8$ and above scales.

Last, considering that remote sensing image SR is often used as a preprocessing step for downstream tasks such as semantic segmentation and object detection, in the future, a joint training framework of FDFNet and downstream tasks will be explored. By introducing semantic constraints of downstream tasks (such as segmentation accuracy-oriented loss terms) into the loss function, the SR results can better meet the feature requirements of downstream tasks, achieving end-to-end performance improvement through multitask collaboration and further expanding the practical application value of the model.

V. Conclusion

In this study, we propose an efficient frequency decoupling-fusion network, termed FDFNet, which effectively handles the wide coverage, rich high-frequency details, and large scene variations inherent in remote sensing images. Unlike previous methods that primarily operate in the spatial domain and often suffer from the loss of high-frequency information, our approach introduces an FDB to explicitly separate low- and high-frequency components. This division allows dedicated processing of each part in a divide-and-conquer strategy, thereby facilitating earlier recovery of fine details. Furthermore, to mitigate the limitation of spatial self-attention, which acts as a low-pass filter and struggles with high-frequency reconstruction, we design a PFTM. This module concurrently incorporates spatial frequency attention and channel transposition attention, enhancing the model's ability to capture local texture details and reconstruct high-frequency information more effectively. Extensive experiments conducted on multiple public remote sensing datasets demonstrate that FDFNet achieves superior performance compared to SOTA methods, both in terms of quantitative metrics and visual quality. In the future, we believe that our method can be easily applied to other image restoration problems, such as image deraining and defogging.

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Kanghui Zhao received the B.S. degree from the School of Computer Science and Engineering, Wuhan Institute of Technology, Wuhan, China, in 2019, and the master's degree from Wuhan Institute of Technology in 2022, where he is currently pursuing the Ph.D. degree with Hubei Key Laboratory of Intelligent Robot, under the supervision of Prof. Tao Lu.

His research field includes image/video processing and computer vision.

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Tao Lu received the B.S. and M.S. degrees from the School of Computer Science and Engineering, Wuhan Institute of Technology, Wuhan, China, in 2003 and 2008, respectively, and the Ph.D. degree from the National Engineering Research Center for Multimedia Software, Wuhan University, Wuhan, in 2013.

He was a Post-Doctoral Researcher at the Department of Electrical and Computer Engineering, Texas A&M University, College Station, TX, USA, from 2015 to 2017. Now, he is an Associate Professor at the School of Computer Science and Engineering, Wuhan Institute of Technology, and a Research Member at Hubei Provincial Key Laboratory of Intelligent Robot. His research interests include image/video processing, computer vision, and artificial intelligence.



Jiaming Wang received the Ph.D. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 2022.

He is currently a Teacher at the School of Computer Science and Engineering, Wuhan Institute of Technology, Wuhan. His research field includes image/video processing and computer vision.



Yu Wang received the master's degree from Wuhan Institute of Technology, Wuhan, China, in 2021. He is currently pursuing the Ph.D. degree with the State Key Laboratory for Information Engineering in Surveying, Mapping, and Remote Sensing, Wuhan University, Wuhan, under the supervision of Prof. Zhenfeng Shao.

His research field includes image/video processing and computer vision.



Yuanzhi Wang received the B.S. and master's degrees from Wuhan Institute of Technology, Wuhan, China, in 2019 and 2022, respectively. He is currently pursuing the Ph.D. degree with the School of Computer Science and Engineering, Nanjing University of Science and Technology, Nanjing, China.

His research interests include multimodal machine learning, generative modeling, AIGC, image/video synthesis, processing, and analysis.



Yanduo Zhang received the Ph.D. degree from Harbin Institute of Technology, Harbin, China, in 1999.

He was a Visiting Professor at the Center for Vision, Cognition, Learning, and Art, University of California at Los Angeles, Los Angeles, CA, USA, from 2012 to 2013. He is currently a Professor with Wuhan Institute of Technology, Wuhan, China. He has co-authored two books and published more than 100 research articles. His research interests include computer vision, robot control, and intelligent computing.

Dr. Zhang is a member of the ACM.

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